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CS 82A - Data Science

module2_lab2.ipynb

Textbook Problem 4.2 -

Multiply with addition and loops in a function

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Do it with a While Loop

```
In [1]: # Problem 4.2
# when multiplying x * y, assignment operators must be done
# result = result + x
first_number = 7
second_number = 6
# initialize these to 0, result uses the augmented assignment
def while_multiply(first_num, second_num):
    count = 0
    result = 0
    while(count < second_num):      # how many times should
        count += 1                  # count the iterations
        result = result + first_num # accumulate the sum
    return result

answer = while_multiply(first_number, second_number)
print("answer is:", answer)
print(first_number, " * ", second_number, " = ", answer)
```

answer is: 42

7 * 6 = 42

How does the while loop terminate? The while loop always iterates if the condition is true. Remember to set a count and increment the count (or index) or it will loop infinitely. In other words "A while

loop needs its counter updated inside the loop, or it runs forever." Set a limit with a comparison operator. Comparison operators return True or False depending on the comparison.
(<https://www.w3schools.com/python/python>

Do it with a For Loop

```
In [2]: # initialize these to 0, result uses the augmented assignment
# To loop through a set of code a specified number of times
# The range() function returns a sequence of numbers, starting from 0 by default
def for_multiply(first_num, second_num):
    count = 0
    result = 0
    for count in range(second_num): # range begins from 0
        # count += 1 # range already increments
        result = result + first_num # accumulate the sum
    return result

answer_2 = for_multiply(first_number, second_number)
print("answer is:", answer_2)
print(first_number, "*", second_number, "=", answer_2)
```

answer is: 42

7 * 6 = 42

This **for loop** uses the `range()` function to increment up to the parameter. The keyword "in" links to the sequence you want to iterate over. The statement is executed each iteration.

End Of Lab - Behzad Ghabaei

In []: